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Authors for correspondence:

Subhendu Bhandary

e-mail: bhandary.subhendu@gmail.com

Partha Sharathi Dutta

e-mail: parthasharathi@iitrpr.ac.in

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Adaptive behavioural strategies promote species persistence in niche-constrained ecological communities

Subhendu Bhandary^{1,2} and Partha Sharathi Dutta²

¹Chair of Network Dynamics, Institute of Theoretical Physics, TUD Dresden University of Technology, Dresden, Germany

²Department of Mathematics, Indian Institute of Technology Ropar, Rupnagar, Punjab, India

SB, 0000-0003-0870-1234; PSD, 0000-0001-6067-1023

Species persistence in ecosystems results from the interplay between ecological interactions and behavioural strategies. Classical niche theory explains how resource overlap determines the strength of interactions, while evolutionary game theory (EGT) describes how cooperation and defection strategies arise from payoff-dependent decisions. Most eco-evolutionary models assume that strategies remain fixed over generations, overlooking the potential for behavioural adjustments on ecological time scales. Here, we develop a mathematical framework that integrates Lotka–Volterra niche competition (LVNC) dynamics with an adaptive strategy model, allowing species to update their strategies based on the payoff-sensitive Fermi rule. We examine community outcomes on four symmetric games: prisoner's dilemma (PD), snowdrift, stag hunt (SH) and trivial (TR) (also known as harmony), implemented on networks with varying topologies and connectivity. We find that adaptive strategy updates significantly reduce the likelihood of species extinction in comparison to static strategies, enabling coexistence in payoff regions that would otherwise lead to collapse. The structure of the network plays an important role in determining persistence, particularly in snowdrift and PD games. Furthermore, regular ecological networks may experience abrupt extinction with changes in connectivity,

whereas scale-free networks support persistence through cooperative hubs. Overall, our results demonstrate that behavioural adaptation is pivotal in promoting species persistence, more so than network architecture alone.

1. Introduction

Species fitness arises from a regular interplay of local interactions among individuals sharing space, competing for resources or benefiting from mutualistic exchanges, together with broader eco-evolutionary processes, such as dispersal, mutation and selection [1,2]. Classical niche theory explains how trait matching and resource-use similarity structure these interactions and determine the degree of coexistence among species [3–5]. Niche overlap can be understood as the extent to which species use similar resources, habitats or environmental conditions [5,6]. Greater overlap increases the potential for species to influence one another, and in the Lotka–Volterra niche model, this idea is represented by converting overlap into pairwise interaction coefficients [7,8]. Species with stronger niche overlap typically experience greater competition or facilitation depending on the nature of their interaction [6,9]. Yet ecological similarity alone does not determine species outcomes: species interactions are often context-dependent, with species shifting among facilitative, competitive or exploitative responses depending on local environmental conditions and neighbouring species [10–12]. For example, alpine plants with overlapping local resource use may compete under benign conditions, but under high environmental stress, neighbouring plants can buffer harsh conditions and facilitate one another [11,12]. Evolutionary game theory (EGT) formalizes these behavioural aspects by assigning payoffs to alternative strategies and describing how such strategies spread in structured populations [13–15]. In dynamic environments, species may revise strategies to maximize fitness through processes akin to strategic optimization or imitation observed in biological and social systems [16–18]. Understanding how such adaptive behavioural responses interact with niche-structured ecological constraints is therefore essential for understanding long-term community persistence.

EGT is widely used to study behavioural adaptation in structured populations and adaptive networks, where individuals update their strategies based on local payoff differences [19–21]. A key result from these studies is that network topology strongly influences cooperative behaviour [22–24], as shown in studies on regular lattices, small-world graphs and scale-free networks [25,26]. Update rules, such as the Fermi process, capture stochastic, payoff-dependent behavioural shifts and have become standard tools for modelling learning and imitation in biological systems [27,28]. However, in most such models, payoffs are defined independent of the ecological similarity between interacting species, or depend on aggregate environmental variables that change with strategy frequencies [24,29]. Even when eco-evolutionary feedbacks are included, behavioural dynamics typically affects payoffs via environmental change, while the niche-based structure of pairwise interactions is not explicitly represented. As a result, behavioural adaptation and niche-based ecological constraints are often treated separately, despite being closely coupled in natural systems.

Parallel to these developments, the classical Lotka–Volterra competition model [30,31] and its niche-based extension, Lotka–Volterra niche competition (LVNC) model, provide a mechanistic description of species interactions in which pairwise interaction coefficients are determined by the degree of overlap between species' resource-use distribution [7,8]. Within this framework, pairwise interaction coefficients are derived from niche similarity, enabling a unified representation of competition, mutualism, predation and antagonistic interactions in a single ecological model. LVNC has therefore been widely used for a variety of empirical and theoretical contexts, ranging from microbial assemblages to plant communities and synthetic ecosystems [32–36]. However, a major limitation of LVNC is that species cannot adjust their behavioural responses even as neighbourhood composition or local environmental conditions change. Although recent attempts have incorporated cooperation–defection strategies into LVNC by embedding game-theoretic

payoffs [37], these approaches still treat payoffs as largely static and decoupled from dynamic strategy change and network heterogeneity. Consequently, the important link between niche-based ecological similarity and adaptive behavioural dynamics remains largely unexplored.

We address this gap by introducing a niche-constrained adaptive game framework. Unlike eco-evolutionary game models, where strategies reshape the environment and thereby modify payoffs, we assume that ecological constraints arising from niche overlap are fixed and externally imposed, setting bounds on the strength of interspecific interactions. Within these limits, behavioural adaptation dynamically reweights these pairwise interactions without changing the underlying niche structure. Importantly, these behavioural updates operate at the level of individual pairwise interactions, rather than as species-level traits, capturing context-dependent behavioural plasticity rather than long-term evolutionary strategy fixation. Ecological constraints are held constant, ensuring that observed dynamical changes originate solely from behavioural adaptation.

Here, we ask how payoff-driven behavioural adaptation operates when constrained by fixed ecological similarity, and how this previously overlooked eco-behavioural coupling shapes long-term community persistence. To address this, we combine interaction-specific behavioural dynamics into the LVNC framework, in which payoff magnitudes are determined by niche overlap while behavioural states update through a payoff-dependent Fermi rule. This formulation explicitly decouples behavioural adaptation from environmental modification, allowing us to examine the stabilizing effects of behavioural flexibility alone. By analysing this eco-behavioural coupling for regular, small-world, random and scale-free networks, each characterized by distinct clustering and degree heterogeneity [38,39], we demonstrate how ecological constraints and adaptive behaviour jointly regulate community persistence. Our results show that adaptive behavioural responses can substantially suppress extinction on diverse interaction regimes, while variations in network structure play a limited role.

2. Methods

(a) The Lotka–Volterra niche competition model

The well-known Lotka–Volterra model of species interaction can be written as [8]

$$\frac{dy_i}{dt} = r_i y_i \left(1 + \frac{\sum_{j=1}^N a_{ij} y_j}{K_i} \right), \quad i = 1, \dots, N, \quad (2.1)$$

where y_i represents the abundance of the i th species, r_i is its intrinsic growth rate and a_{ij} is the competition coefficient quantifying the effect of species j on species i . Specifically, for $j \neq i$, a_{ij} denotes interspecific competition, while $a_{ii} = -1$ accounts for intraspecific competition, which is always negative to signify the competitive nature of interactions; K_i is the carrying capacity of species i and N is the total number of species in the community. Here, we assume $r_i = r$ and $K_i = K$ for all species. Since r is common to all species, it is absorbed into the time variable during non-dimensionalization. By normalizing species abundance as $x_i = y_i/K$, equation (2.1) simplifies to

$$\frac{dx_i}{dt} = x_i \left(1 - x_i + \sum_{j=1, j \neq i}^N a_{ij} x_j \right), \quad i = 1, \dots, N. \quad (2.2)$$

This model (equation (2.2)) forms the basis for analysing competition dynamics within ecological communities.

Competition arises because species exploit overlapping sets of limited resources, such as food, habitat or other ecological factors. Niche theory [7] provides a mechanistic basis to understand this competition through resource-utilization functions along a niche axis. This axis represents a continuous gradient of resource availability, such as prey size or habitat preferences. Each i th species'

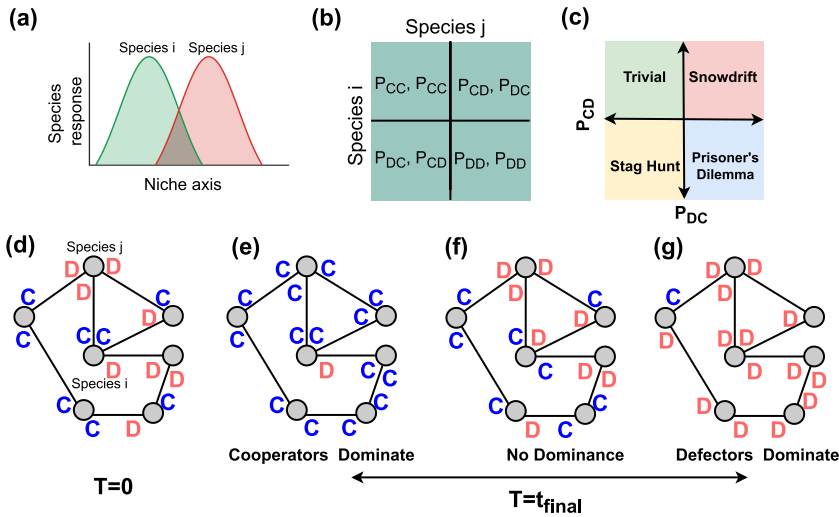


Figure 1. Schematic overview of niche-constrained and payoff-driven species dynamics. The top row illustrates: (a) Gaussian niche-overlap interactions, (b) the payoff matrix between species i and j and (c) how different game types appear as the payoff parameters P_{CD} and P_{DC} vary. The bottom row presents: (d) a species interaction network in which nodes initially adopt cooperative (C) or defective (D) strategies and update them using the strategic update rule, leading to (e) a cooperator dominance state, (f) a stable mixed state or (g) a defector dominance state.

niche is defined by its optimal resource utilization (η_i), and the degree of resource distribution is determined by the niche width (σ_i). Resource utilization along the niche axis is described by a bell-shaped function [7]:

$$H_i(\eta) = e^{-\frac{(\eta - \eta_i)^2}{2\sigma_i^2}}, \quad (2.3)$$

which quantifies the suitability of different resource levels η for species i , centred at the optimum niche value η_i with a variance σ_i^2 determining the width of the curve (figure 1a). In our simulations, we fix the number of species N , and assign each species a niche centre $\eta_i \in [0, 1]$, drawn uniformly at random to ensure coverage over the niche axis and realistic heterogeneity in resource preference. Unless otherwise stated, we assume a uniform niche width $\sigma_i = \sigma = 0.15$ for all species, using this as a representative intermediate niche width; results for additional niche widths are shown in fig. S3 in the supplementary material.

With niche centres $\eta_i \in [0, 1]$, the niche-overlap weight between species i and j , denoted H_{ij} , is determined by the overlap between their normalized resource-utilization functions:

$$H_{ij} = -\frac{\int H_i(\eta)H_j(\eta)d\eta}{\int H_i^2(\eta)d\eta} = -e^{-\frac{(\eta_i - \eta_j)^2}{4\sigma^2}}. \quad (2.4)$$

After incorporating the niche overlap equation (2.4) into the model (equation (2.2)), the interaction coefficient a_{ij} becomes

$$a_{ij} = \theta_{ij}H_{ij}, \quad (2.5)$$

where θ_{ij} is an element of the adjacency matrix $A = [\theta_{ij}]$ that specifies whether species i interacts with species j ($\theta_{ij} = 1$) or not ($\theta_{ij} = 0$). In the corresponding species interaction network, the degree of the i th node, k_i , is determined by θ_{ij} , and is defined as the number of neighbours of node i :

$$k_i = \sum_j \theta_{ij} \quad (2.6)$$

and the average degree of the network, k , is given by

$$k = \frac{\sum_i k_i}{N}. \quad (2.7)$$

Modifying the Lotka–Volterra competition model (equation (2.2)), where a_{ij} follows the niche-overlap formulation (equation (2.5)), yields the LVNC model. To ensure generality of our results, we analyse this model for multiple network architectures: regular, small-world, random and scale-free networks [25,38,40,41]. Regular networks were generated as k -regular ring lattices with $k/2$ neighbours on each side, small-world networks were generated from the same ring lattice using Watts–Strogatz rewiring probability $p = 0.5$, random networks were generated as Erdős–Rényi graphs with edge probability $p_{ER} = k/(N - 1)$ and scale-free networks were generated using preferential attachment with attachment size $m = k/2$, starting from a complete initial core of size $m + 1$. Furthermore, we examine two community sizes ($N = 101$ and $N = 1000$) and systematically vary network connectivity by changing the average degree. Results for $N = 101$ with $k = 4$ correspond to random networks shown in figures 2–5 and in figs. S1–S6 in the supplementary material, while results for $N = 1000$ correspond to figure 6 and in figs. S7–S10 in the supplementary material. The realized properties of the generated interaction networks for $N = 1000$, including the average degree and clustering coefficient for each topology, are summarized in table S1 in the supplementary material.

(b) Integrating niche overlap and species strategic interactions via dilemma games

While the LVNC framework effectively captures the strength of interspecific interactions based on ecological similarity (e.g. niche overlap), it does not account for the behavioural dimension of species interactions. In natural ecosystems, species not only experience interaction intensities but also make context-dependent decisions—such as cooperating, competing or exploiting—based on environmental and social cues [37]. To incorporate this behavioural flexibility, we embed EGT into the LVNC framework, treating interspecies interactions as strategic encounters.

In this extended framework, each species adopts a pure strategy: either cooperation (C) or defection (D), when interacting with another species [42]. Ecologically, cooperative interactions represent reduced competitive pressure or facilitative effects, while defective interactions represent stronger competitive or exploitative effects; in resource-use terms, this corresponds to less versus more aggressive resource use. The resulting species fitness owing to these strategic interactions is not fixed but depends on the outcome of pairwise interactions. These outcomes are formalized using a payoff matrix P , which quantifies the fitness benefit or cost a species receives from a given interaction. Specifically, for a pair of interacting species i and j , the payoff matrix is

$$P = [P_{ij}] = \begin{array}{c} \text{Species-}i \\ \begin{array}{cc} C & D \\ D & \end{array} \end{array} \begin{array}{c} \text{Species-}j \\ \begin{array}{cc} C & D \\ \hline \begin{array}{cc} P_{CC} & P_{CD} \\ P_{DC} & P_{DD} \end{array} \end{array} \end{array}$$

where P_{CC} , P_{CD} , P_{DC} and P_{DD} are the payoffs for the i th species (figure 1b). The corresponding payoffs for the j th species are given by the transpose of P . Here, we consider symmetric games.

The ranking of payoff values P_{ij} determines diverse social dilemmas (figure 1b,c). For instance, in the prisoner's dilemma (PD) game, defection dominates, making cooperation difficult to sustain, with payoffs ranked $P_{CD} < P_{DD} < P_{CC} < P_{DC}$ [43]. In the snowdrift (SD) game, partial cooperation appears, with the payoffs ranked as $P_{DD} < P_{CD} < P_{CC} < P_{DC}$ [13,43]. The stag hunt (SH) game favours mutual cooperation, with payoffs ranked as $P_{CD} < P_{DD} < P_{DC} < P_{CC}$, reflecting the benefit of coordinated behaviour [44]. Finally, the trivial (TR) game (also referred to as the non-dilemma harmony game in EGT) always favours cooperation, with $P_{CC} > P_{DC}$ and $P_{CD} > P_{DD}$.

[45,46]. Within this EGT-LVNC framework, we consider a species' strategy to be either static or adaptive (varies over time). The details of static versus adaptive strategic behaviour are discussed in the following subsections.

(i) Static strategies in species interactions

To incorporate the effects of strategic interactions on community dynamics, we modify the interaction coefficient a_{ij} as given in equation (2.5) of the LVNC model. The modified coefficient, denoted as a'_{ij} , is given by [37]

$$a'_{ij} = P_{ij} \times |a_{ij}|, \quad (2.8)$$

where P_{ij} is chosen from the set $\{P_{CC}, P_{CD}, P_{DC}, P_{DD}\}$ depending on the pairwise strategic relationship between the species. For instance, $P_{ij} = P_{CD}$ when the i th species cooperates and the j th species defects. Thus, $|a_{ij}|$ determines the magnitude of the niche-constrained interaction, whereas the payoff P_{ij} determines the sign of a'_{ij} . Consequently, a'_{ij} can take positive, negative or zero values, depending on the payoff values. A negative a'_{ij} indicates a competitive effect of species j on species i , while a positive a'_{ij} indicates a facilitative effect. Pairwise competition, mutual facilitation or asymmetric exploitation then arise from the combined signs of a'_{ij} and a'_{ji} . Following previous work [37], we fix the payoff values at $P_{CC} = 0.05$ and $P_{DD} = -1$, and explore mixed-interaction payoffs within the ranges $P_{CD} \in [-2, 0]$ and $P_{DC} \in [-1, 1]$ to ensure ecological relevance. To examine the sensitivity of the model to the choice of payoff values, we additionally consider moderate shifts in the payoff matrix together with payoff regimes favouring stronger cooperative effects (supplementary material, fig. S4). To study community dynamics under static strategies, among all active links present in a network, a fraction f_C of interactions is randomly chosen as cooperation, and the remaining are defection. Unless stated otherwise, we set $f_C = 0.5$ corresponding to an initially equal proportion of cooperative and defective interactions.

(ii) Adaptive strategies in species interactions

To investigate community dynamics when species strategies vary over time, we consider that each species interacts with its neighbours in a network and that its strategy is influenced by the payoffs accumulated from these interactions. Although the underlying ecological niche overlaps $|a_{ij}|$ remain fixed, species can adjust their behaviour dynamically in response to the strategies of neighbouring species, leading to time-dependent payoffs $P_{ij}(t)$. This represents ecologically observed behavioural plasticity, whereby species respond to local interaction outcomes without changes in their fundamental niche or resource availability. Initially, each species is randomly assigned to one of the two strategies, either C or D. Over time, strategies may change as species imitate the most successful neighbours according to the Fermi update rule [28,47]. Details of this updating mechanism are provided in the following paragraphs.

Each species i maintains a strategy vector whose length equals its degree k_i (equation (2.6)), representing the strategies employed in its interactions with each neighbour. Thus, a species may simultaneously cooperate with some neighbours while defecting with others, allowing mixed behavioural responses and reflecting context-dependent plasticity. Thus, $k_i = k_i^C + k_i^D$, where k_i^C and k_i^D denote the number of cooperative and defective interactions of the i th species, respectively. These quantities evolve over time as strategies change. The average payoff of species- i , denoted $\hat{P}_i$, is then given by

$$\hat{P}_i = \frac{1}{k_i} \sum_{j \in \Lambda_i} P_{ij}, \quad (2.9)$$

where Λ_i denotes the set of neighbours of species- i , and P_{ij} is the payoff resulting from the interaction between species- i and species- j . Each species- i may update its strategy by adopting the one used by the neighbouring species with the highest average payoff. Specifically, after calculating payoffs based on current strategies, species- i evaluates each of its neighbours' payoffs and adjusts accordingly. The resulting payoff change ($\Delta\hat{P}_i$) is given by

$$\Delta\hat{P}_i = \hat{P}_i^s - \hat{P}_i, \quad (2.10)$$

where $\hat{P}_i^s = \max_{j \in \Lambda_i} (\hat{P}_j)$ is the maximum average payoff among the neighbours of species i , where $j = s$ is the neighbour that provides the maximum average payoff. Species i then imitates the strategy employed by neighbour s with probability p that follows the Fermi distribution function [28,47]:

$$p = \frac{1}{1 + e^{-\beta\Delta\hat{P}_i}}, \quad (2.11)$$

where β controls the intensity of selection and thereby the degree of randomness in the updating process. At each strategy-update step, species are updated sequentially. The strategy adopted from the highest-payoff neighbour is applied independently to each outgoing interaction link (i, j) , so strategies remain link-specific within the interaction matrix. Here, the Fermi rule is interpreted as a probabilistic fitness-dependent adjustment of interaction behaviour. Importantly, if species i already employs the same strategy as its most successful neighbour along a given connection, its strategy remains unchanged. In our simulations, we set $\beta = 0.1$ unless otherwise specified. This adaptive strategy-updating process and its possible outcomes are illustrated schematically in figure 1d–g.

This link-specific, dynamic behavioural updating makes the effective interaction coefficient a'_{ij} in equation (2.8) time-dependent:

$$a'_{ij}(t) = P_{ij}(t) \times |a_{ij}|, \quad (2.12)$$

where $P_{ij}(t)$ denotes the time-dependent payoff. Therefore, the resulting adaptive LVNC model becomes

$$\frac{dx_i}{dt} = x_i \left(1 - x_i + \sum_{j=1, j \neq i}^N a'_{ij}(t)x_j \right), \quad i = 1, \dots, N, \quad (2.13)$$

thereby incorporating temporally evolving strategies directly into species interactions.

At the individual level, the cooperation level of species i at time t , denoted as $\rho_i^C(t)$, is defined as

$$\rho_i^C(t) = \frac{k_i^C(t)}{k_i}, \quad (2.14)$$

where $k_i^C(t)$ is the number of cooperative interactions of species i at time t . At the community level, the mean cooperation level on the network is given by

$$\rho^C(t) = \frac{1}{N} \sum_i \rho_i^C(t), \quad (2.15)$$

which quantifies the overall effect of strategy changes over the entire network. The corresponding defection fraction is defined as $\rho^D = 1 - \rho^C$.

(c) Equilibrium dynamics and ecosystem resilience

To analyse the ecosystem dynamics under static strategies, we first compute the equilibrium state x_i^* (by substituting equation (2.8) into equation (2.2) and solving for the steady state). From these equilibrium values, we evaluate two complementary community-level indicators: the relative yield total (RYT), which represents the total community abundance, and the probability of species extinction (PSE), which quantifies the fraction of species that fall below a viability threshold. Together, these metrics provide insights into the distribution of species abundances and the vulnerability of species to extinction, respectively.

Once the system reaches equilibrium, the RYT is defined as the total species abundance in the ecosystem and can be determined as

$$\text{RYT} = X = \sum_{i=1}^N x_i^*, \quad (2.16)$$

where x_i^* is the equilibrium abundance of species i and N is the total number of species.

The PSE measures the proportion of species whose abundances fall below a specified threshold, indicating an increased likelihood of extinction. Mathematically, PSE is calculated as

$$\text{PSE} = \frac{N_{\text{extinct}}}{N}, \quad (2.17)$$

where N_{extinct} is the number of species whose abundances fall below the lower threshold (i.e. $x_i < 10^{-2}$). Lower PSE values indicate greater community persistence and stability, whereas higher values signal increasing ecological fragility and an increased risk of collapse.

All simulations were performed in Julia version 1.4.2 [48], and data visualization was performed in R version 3.6.2 [49].

3. Results

(a) Role of static behavioural strategy in species persistence

For static strategic interactions, figure 2a,b shows RYT and PSE over four game types: PD, SD, SH and TR, obtained by varying payoff values P_{CD} and P_{DC} while keeping P_{CC} and P_{DD} fixed. For all games in the payoff plane, RYT increases when moving from PD or TR toward SD, where it attains its maximum, whereas SH consistently remains the lowest RYT (figure 2a). PSE exhibits the opposite pattern: SD supports the lowest extinction and SH the highest (figure 2b), indicating that SD promotes greater species persistence. This higher RYT in SD arises because, under static mixed strategies, cooperative-defective interactions receive relatively favourable payoffs, whereas mutual defection remains strongly penalized, reducing the average competitive pressure across the community. For the static-strategy simulations under the present parameter setting, extinction is not completely eliminated, as PSE remains non-zero across the payoff regimes shown in figure 2b.

Averaging these metrics over each payoff region (figure 2c,d) further highlights the ordering of community outcomes. Mean RYT follows $\text{PD} \approx \text{SH} < \text{TR} < \text{SD}$, while mean PSE follows $\text{SH} > \text{PD} > \text{TR} > \text{SD}$. These patterns arise from the underlying payoff structures: in SD, unilateral cooperation yields substantial benefits, leading to high community abundance and reduced extinction risk. By contrast, PD and SH impose stronger penalties for failed coordination or exploitation, thereby suppressing productivity and persistence. Taken together, these results demonstrate that the interplay between payoff structure and strategic interactions drives community productivity and species persistence under static strategies, with anti-coordination dynamics in SD producing the most favourable ecological conditions for coexistence.

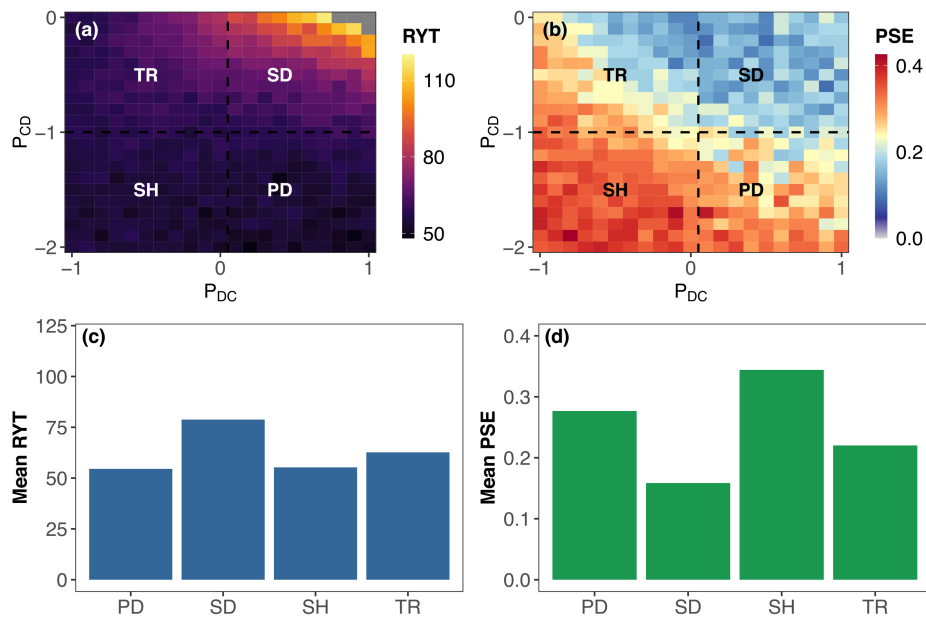


Figure 2. Ecosystem dynamics under static strategic interactions and diverse payoff structures. (a) RYT and (b) PSE on the payoff plane containing all four different game types obtained by varying P_{CD} and P_{DC} . Grey tiles indicate RYT values greater than 120. Dashed lines separate the payoff plane according to different games, for fixed values $P_{CC} = 0.05$ and $P_{DD} = -1$. Transitions from PD (or TR) games toward the SD game are associated with higher RYT, reflecting increased total community abundance; hence, SD games consistently show lower PSE than other payoff regions. (c) Mean RYT averaged over each payoff region, revealing the ordering $PD \approx SH < TR < SD$. (d) Mean PSE averaged over the same regions, following the reverse trend $SH > PD > TR > SD$. Overall, under static strategies, the SD game yields the highest persistence (lowest PSE) and the SH game exhibits the lowest persistence (highest PSE) of species.

(b) Effects of adaptive strategy updates on species persistence

Extending the analysis beyond static strategies, we next examine how adaptive behavioural updates influence ecosystem dynamics by allowing species to change strategies using the Fermi rule. In this framework, species modify their behaviour in response to local payoffs and interactions, generating feedback on the network that influences the outcomes of others. This feedback produces heterogeneous spatial organization, with localized clusters of cooperation or defection that shape variation in species abundance and persistence.

Figure 3a,b presents RYT and PSE for adaptive strategy updates over payoff regions with different game types. TR and SH interactions consistently yield high RYT and low extinction risk, indicating stable cooperative regimes under adaptive dynamics. SD games also sustain high productivity and persistence, although moderate variability arises owing to their anti-coordination payoff structure. By contrast, the PD results in more heterogeneous outcomes, with points A, B and C in figure 3a indicating regions of high, intermediate and low RYT, respectively, within the same payoff region. Averaging outcomes for all payoff regions, figure 3c depicts that mean RYT follows $PD < SD < SH \approx TR$, and figure 3d shows mean PSE is highest in PD while remaining negligible in the other game types. Thus, extinction under adaptive strategy is largely restricted to narrow regions of PD payoff space. These results arise because adaptive updates continuously rescale interaction strengths in response to realized payoffs, allowing particular strategic outcomes to dominate locally. As a result, cooperation becomes increasingly prevalent in coordination- and cooperation-favouring games, stabilizing species densities, whereas PD adaptive dynamics generate coexistence of contrasting local states. These qualitative patterns are robust to changes in

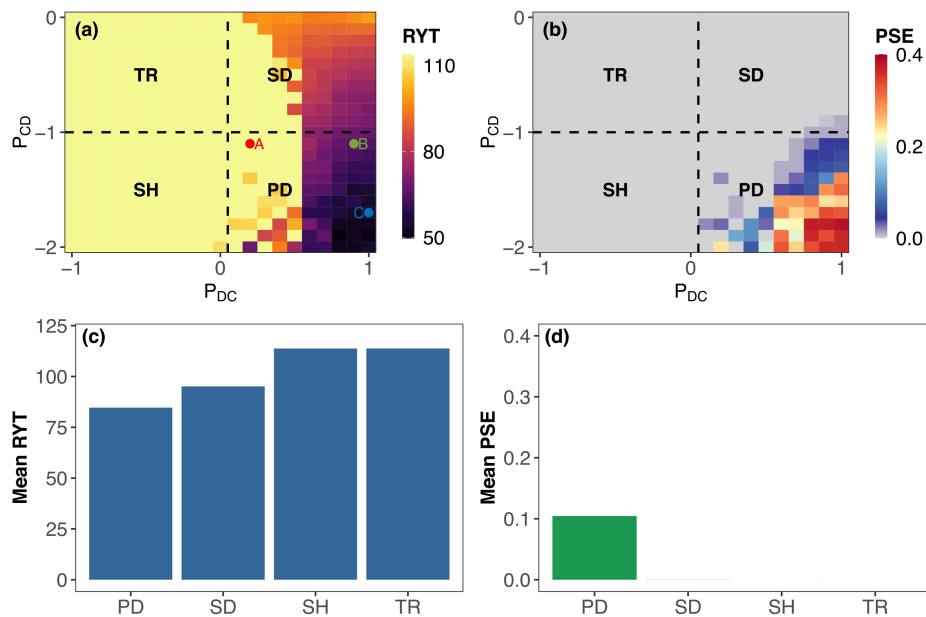


Figure 3. Ecosystem dynamics under adaptive strategic interactions. (a) RYT and (b) PSE are shown for different games in the payoff plane, illustrating the effects of adaptive strategies implemented via the Fermi update rule. Points A, B and C, denoted in sub-figure (a), represent high, intermediate and low RYT within the PD region, highlighting the diversity of outcomes compared with other games where RYT is consistently high. (c) Bar plot of mean RYT, and (d) bar plot of mean PSE calculated over all payoff regions summarize overall patterns of species persistence. Most species survive under TR, SH and SD games, while extinction is limited to small regions in the PD game. Mean RYT follows $PD < SD < SH \approx TR$, and mean PSE is non-zero only for the PD game. Overall, these results indicate that adaptive strategies increase species persistence and maintain high community productivity across a wide range of game scenarios, except the PD game.

initial cooperation levels, selection intensities and niche widths, although strongly defector-biased initial conditions reduce coexistence, particularly in the SH and PD regions (see supplementary material, figs. S1–S4). Together, these results show that adaptive strategies substantially increase species persistence and maintain high productivity over most payoff regions, while simultaneously introducing dynamical heterogeneity in conflict-prone environments.

To elucidate the origin of this variability within the PD region, we examine three representative payoff points (denoted as A, B and C) corresponding to high, intermediate and low community yield, respectively. Species-level dynamics at these points are shown in figure 4. At point A (figure 4a,d), all species converge to stable, non-zero abundances, indicating complete persistence. Point B (figure 4b,e) exhibits sustained oscillations in species densities, resulting in bounded variability around a quasi-stationary state. At point C (figure 4c,f), most species collapse to extinction, with only a few persisting at low abundance. These patterns demonstrate that adaptive strategies can generate markedly different ecological states within the same payoff region by reshaping effective interaction strengths and altering extinction risk.

To understand the underlying behavioural mechanisms, we track, for each species, the fraction of its interactions with neighbours that are cooperative versus defective at each time point (figure 5). Averaging across species, these trajectories reveal the dominant strategy at each point. At point A, cooperation prevails over all species, supporting stable high-density coexistence (figure 5a,d). At point B, the fraction of cooperators fluctuates continuously, reflecting dynamic alternation between cooperation and defection; species densities do not reach a fixed equilibrium but fluctuate persistently around a quasi-stationary state (figure 5b,e). At point C, defection dominates, limiting cooperative interactions and increasing extinction probability (figure 5c,f). These

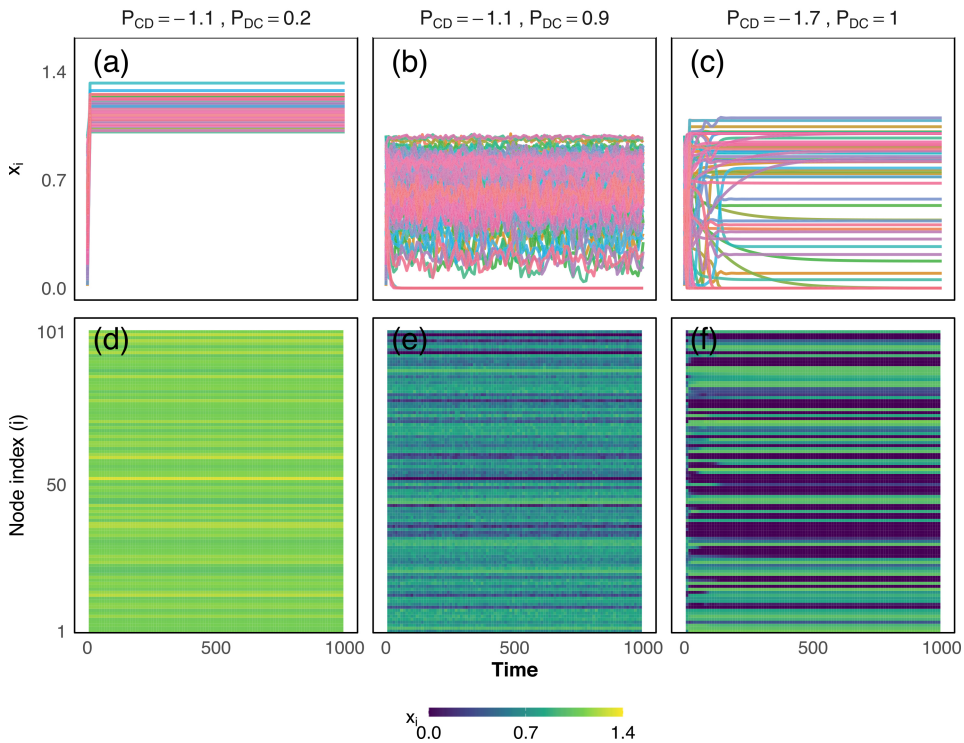


Figure 4. Spatiotemporal species dynamics under adaptive strategy updating. Panels (a)–(c) show the time series of species abundances for the three representative payoff points A, B and C, respectively, marked in figure 3. Panels (d)–(f) display the corresponding spatiotemporal density maps, with time on the horizontal axis, species index on the vertical axis, and colour bar indicating abundance. At point A, all species converge to a stable steady state with non-zero densities. At point B, species abundances do not settle into a fixed equilibrium but instead exhibit sustained, bounded fluctuations around a quasi-stationary state. At point C, some species persist at low abundance, while a large fraction of the species goes extinct. Adaptive strategy updating produces distinct persistence outcomes depending on payoff values.

analyses reveal that adaptive behavioural updates not only increase persistence but also generate spatially and temporally heterogeneous patterns in conflict-prone games. Adaptive strategies, therefore, support multiple coexistence regimes while maintaining community productivity, particularly in coordination-favouring scenarios such as TR and SH.

(c) Role of network topology in determining species persistence

To assess whether persistence patterns depend on interaction network structure under static (supplementary material, figure S9) and adaptive (figure 6; and supplementary material, figs. S7, S8 and S10) strategies, we examine species extinction for four widely studied network topologies: regular, small-world, random and scale-free, each comprising $N = 1000$ species and three levels of average connectivity ($k = 4, 6$ and 8). Figure 6 shows the resulting PSE on the payoff plane for each case. For all network types, adaptive updating strongly suppresses extinction in the TR and SH games, demonstrating that the advantages of dynamic strategy adjustment are broadly robust to structural variation.

Network topology-dependent differences in PSE arise primarily in the SD and PD regimes, where cooperation is most fragile. For $k = 4$, regular networks exhibit the lowest PSE, suggesting that homogeneous low-degree neighbourhoods initially buffer species against defection-driven collapse. However, as connectivity increases, regular networks undergo a sharp transition (supplementary material, fig. S8) in the PD region: strong local clustering allows defector strategies to

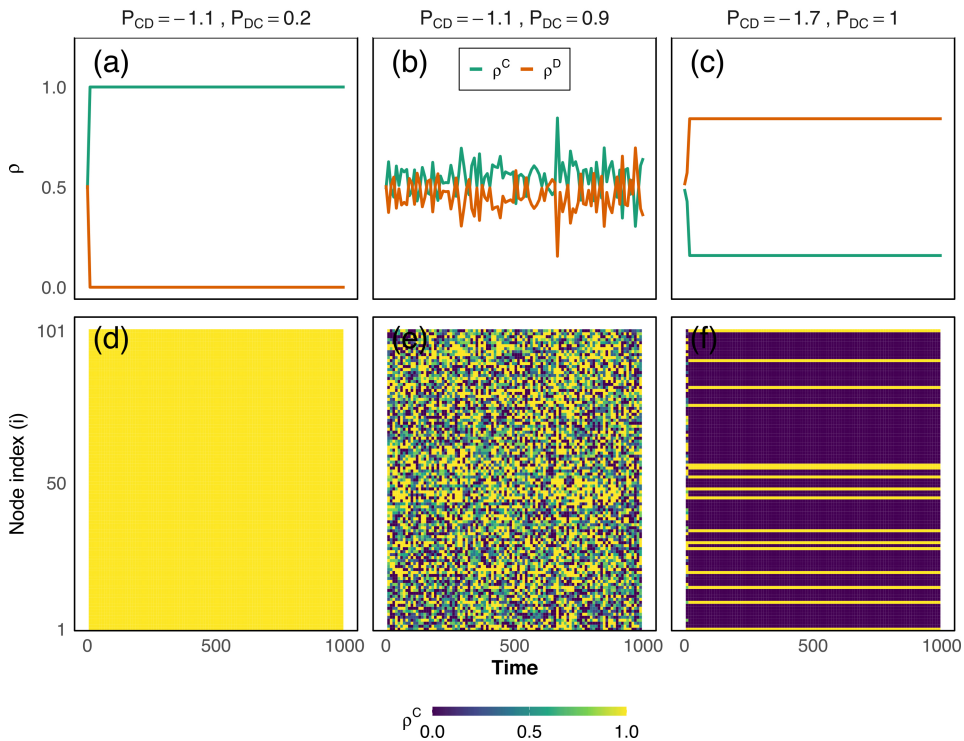


Figure 5. Species-level cooperation dynamics under adaptive strategy updates. Panels (a)–(c) show the mean fraction of cooperation (ρ^C) and defection (ρ^D) at each time step, computed by averaging the fraction of cooperative interactions for all species for the three representative payoff points *A*, *B* and *C* marked in figure 3. Colours indicate the cooperation fraction ρ^C and defection fraction $\rho^D = 1 - \rho^C$, as shown in the legend. Panels (d)–(f) present the corresponding spatiotemporal distributions of individual species, with time on the horizontal axis, species index on the vertical axis and colour indicating the fraction of cooperation of each species toward its neighbours. At point *A*, cooperation dominates, and all species maintain high cooperative interactions. At point *B*, cooperation and defection fluctuate dynamically, with no single strategy prevailing. At point *C*, defection dominates in most species, while a few maintain cooperative links. This figure illustrates how adaptive strategy updates generate distinct behavioural regimes, shaping overall patterns of community cooperation over time.

spread rapidly, leading to a sudden increase in the fraction of extinct species. By contrast, small-world, random and scale-free networks show more gradual changes in extinction risk with increasing connectivity (supplementary material, figs. S8 and S10). Among these networks, scale-free networks consistently attain the lowest PSE for high k , as heterogeneous degree distributions generate cooperative hubs surrounded by many low-degree nodes, enabling local persistence even when defection becomes globally advantageous. Random and small-world networks, both characterized by lower clustering at comparable k , show intermediate responses with smoother transitions between cooperative and defective regions.

Overall, changes in global network topology produce only moderate differences relative to the dominant effects of payoff structure and adaptive imitation. Increasing connectivity generally intensifies competitive interactions by expanding the number of strategic encounters and enlarging extinction-prone regions in the PD and SD games; yet aside from the sharp shift observed in highly connected regular networks, the qualitative structure of extinction landscapes remains largely consistent. These results indicate that while clustering and degree heterogeneity modulate the spread of strategies, species persistence is primarily governed by the interaction between the payoff structure and adaptive behavioural updating.

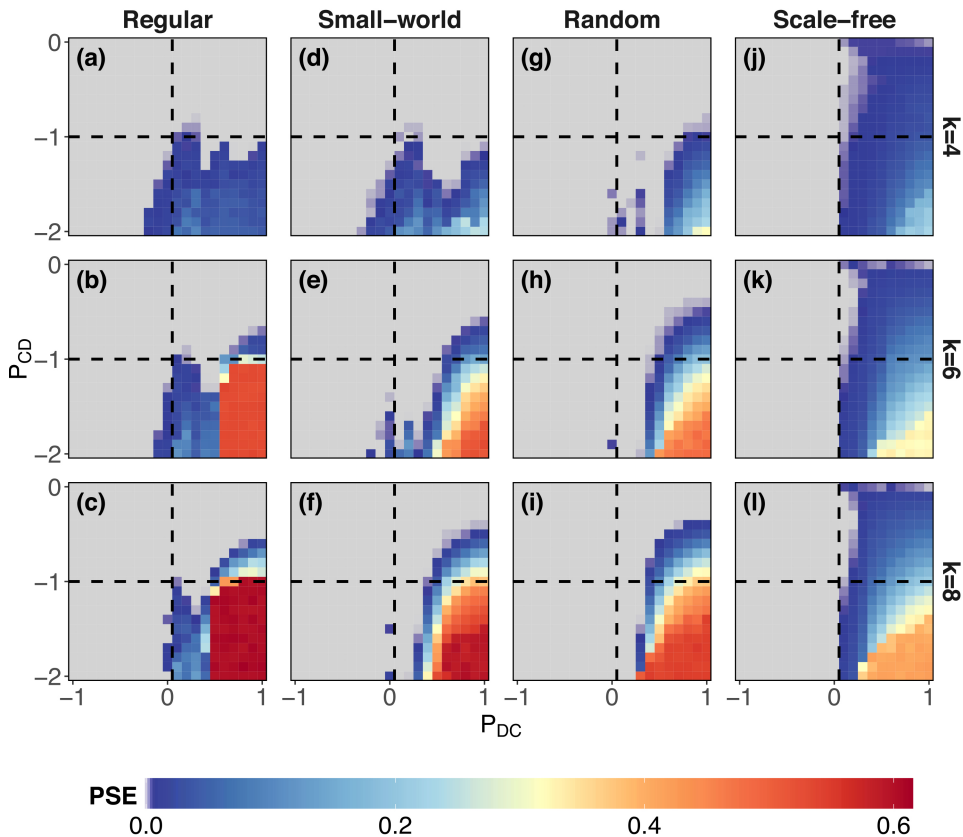


Figure 6. Influence of network topology, connectivity and game payoffs on species extinction under adaptive strategies. PSE is shown for all four network types: regular, small-world, random and scale-free, for three average degrees ($k = 4, 6$ and 8). Network structure has a limited effect, while a higher average degree modestly increases PSE by promoting cluster formation. Game payoffs strongly shape outcomes, with high-defection games (e.g. PD and SD) causing greater species loss. At higher connectivity, in localized regions, extinction follows: regular > small-world > random > scale-free. These results highlight that, although global network structure plays a limited role, connectivity and strategic interactions can locally amplify extinction.

4. Conclusions and discussion

Recent efforts to integrate niche-based ecology with behavioural decision-making [5,37] suggest that species interactions are shaped not only by resource overlap, but also by how individuals adjust their behaviour when conditions change. In nature, such behavioural shifts, from competing to cooperating or vice versa, are triggered by payoff advantages, stress responses or neighbourhood composition [11]. This implies that the persistence of a community cannot be inferred solely from ecological similarity that arises owing to overlapping niches. It also depends on how species update their strategies, which payoff structure governs the incentives to cooperate or defect, and how these interactions propagate through the underlying network topology. Yet these three elements, behavioural adaptation, payoff incentives and structural heterogeneity, are typically examined in isolation. Here, we address this gap by embedding payoff-driven strategy updates into a niche-based Lotka–Volterra model and systematically examining how adaptive behaviour, game type and network topology collectively shape long-term species persistence.

The present model captures key features of ecological systems where interaction strength is constrained by niche similarity, but behavioural responses vary over partners. In microbial communities, metabolically similar strains often compete for shared resources yet may engage in

cross-feeding depending on local payoffs and partner identity [50–52]. In plant communities, neighbours frequently shift from competition to facilitation under environmental stress, with interaction outcomes determined by neighbour identity and local conditions rather than changes in resource-use traits [11,53]. Similarly, hosts can adjust contact or avoidance behaviour toward specific neighbours in response to perceived infection risk, dynamically modulating interaction strength without restructuring ecological niches [54,55]. These examples highlight how payoff-dependent, interaction-specific behavioural responses can dynamically reweight otherwise fixed ecological interaction networks, consistent with the niche-constrained adaptive framework developed here.

Overall, our results show that allowing species to adjust their strategies through the Fermi update rule generally increases persistence relative to fixed behavioural strategies, although the magnitude depends strongly on the payoff structure. In cooperation-favouring games, such as trivial interactions or coordination-based SH games, adaptive behaviour drives convergence to high-abundance, stable states (figure 3), reinforcing coexistence through reduced competitive pressure and persistent mutualistic gains. These outcomes align with eco-evolutionary work, which shows that when strategy dynamics interacts with ecological feedbacks, SH and other cooperation-favouring games can maintain bistable or stable cooperative equilibria [56,57]. In the PD game, defection dominates, even though mutual cooperation would yield higher collective payoffs. Under these conditions, adaptive strategies can delay collapse only when cooperators locally invade and form protective clusters, in line with previous work showing that eco-evolutionary feedbacks only partially rescue cooperation in conflict-dominated games [58]. Correspondingly, we identify three qualitatively distinct dynamical regimes in the PD payoff plane, with adaptive updates stabilizing high, intermediate or low-abundance states depending on the strength of cooperative benefits (figure 4), robust for initial cooperation levels (supplementary material, fig. S1). Selection intensity (β) most strongly affects PD and SD regimes, where stronger selection expands low-density domains (supplementary material, fig. S2). Niche width (σ) further modulates these dynamics: broader niches increase overlap among species and thereby strengthen interaction connectivity across the community. This can support cooperation in coordination games but simultaneously facilitate the spread of defectors in conflict-prone games, thereby increasing extinction risk (supplementary material, fig. S3). Consistent with this, the stability landscape remains largely unchanged with increasing niche width, indicating that behavioural incentives encoded in game payoffs, rather than niche overlap, play the dominant role in shaping community stability (supplementary material, fig. S6). Together, these findings demonstrate that adaptive behaviour reliably promotes coexistence in cooperation-aligned games, while generating more variable outcomes in antagonistic interactions, and these effects hold for all network architectures.

The influence of behavioural dynamics becomes clearer when viewed alongside the effects of species interaction topology (figure 6; supplementary material, figs. S7–S10). Under adaptive strategies, network architecture exerts only a modest global influence, yet connectivity and heterogeneity regulate how cooperation and species abundance are distributed for the system. Highly connected homogeneous networks dilute cooperative gains, weakening the spread of cooperation and reducing total yield (supplementary material, fig. S7). They can also undergo sharp collapses in defection-prone regions (supplementary material, fig. S8). By contrast, heterogeneous networks, especially scale-free structures, provide highly connected hubs that more effectively accumulate and stabilize cooperative strategies. Hence, these hubs increase community abundance, maintain higher cooperation levels, and buffer against defection-driven losses under adaptive strategies (supplementary material, figs. S7 and S10). These structural effects stand in sharp contrast to the static-strategy case (supplementary material, fig. S9), where adaptation is absent. Under static behaviour, extinction risk is consistently higher: no network architecture, payoff regime or connectivity level yields complete persistence. Regular networks experience the greatest species loss, followed by small-world, random and scale-free topologies. Comparing static (supplementary material, fig. S9) and adaptive (figure 6) cases reveals a clear shift: in static systems, persistence is structure-driven, whereas under adaptive behaviour it becomes incentive-driven. These results

reinforce classic insights into the role of topology in promoting cooperation [22,24,59], while demonstrating that their ecological consequences depend critically on whether strategies evolve or remain fixed.

Although our framework captures the interplay between adaptive strategies, interaction games and network topology, some limitations suggest future research directions. First, adaptation is assumed to be cost-free, whereas real organisms face energetic, cognitive or ecological trade-offs. Integrating such costs could clarify when adaptive behaviour stabilizes and when it destabilizes community dynamics [60]. Second, interaction strengths and connections remain static, despite growing evidence suggesting that ecological networks rewire over ecological and evolutionary time scales. Incorporating multiple, distinct time scales for strategy updates and network rewiring may reveal new dynamical regimes and produce more realistic predictions for community persistence [61]. Third, although the Lotka–Volterra formulation captures basic mechanisms of population regulation, it necessarily abstracts away many ecological processes, such as explicit resource dynamics, trait-mediated interactions and epidemiological feedbacks. Embedding adaptive strategies in more mechanistic system-specific models would test whether the dominance of behavioural incentives over network structure persists broadly or not. This perspective is also consistent with recent arguments that ecological game-theoretic models should be coupled to explicit population dynamics, since payoff-based replicator dynamics alone may overlook important ecological constraints on species interactions [62]. Although the present results are robust to the tested ecological-behavioural time-scale ratios (supplementary material, fig. S5), slower behavioural updating produces a modest increase in extinction risk in a limited region of the payoff space. Relative time scales may generate richer dynamics in extended models where ecological conditions, payoff values or network links also change over time [56,57,63]. Finally, many empirical systems, including infectious disease-spreading networks [64] and plant–pollinator networks [65], exhibit both adaptive behaviour and dynamically changing interaction structure. Applying our framework to such empirical datasets offers an avenue to validate the mechanisms identified here.

Overall, our findings suggest that adaptive behavioural strategies can significantly enhance community persistence, particularly in environments experiencing stress or declining habitat quality. This adaptive behavioural flexibility mirrors patterns observed in natural communities, where organisms frequently shift between competition and facilitation depending on resource scarcity, crowding or other stressors [11]. By coupling niche structure with payoff-dependent strategy revision, our model identifies behavioural plasticity as a stabilizing mechanism that complements established coexistence processes, including stabilizing niche differences and fitness equalization [9,66], as well as trait variation and ecological complexity emphasized in biodiversity–ecosystem functioning research [67]. This eco-behavioural feedback helps explain why some real-world plant, microbial and mutualistic networks remain resilient even when interaction strengths weaken or environments deteriorate. More broadly, our results suggest that predictive models of species persistence should explicitly incorporate adaptive decision-making to better capture the behavioural capacity upon which real ecosystems depend.

Data accessibility. This article has no additional data.

Supplementary material is available online [68].

Declaration of AI use. We have not used AI-assisted technologies in creating this article.

Authors' contributions. S.B.: conceptualization, data curation, formal analysis, investigation, methodology, software, validation, visualization, writing—original draft, writing—review and editing; P.S.D.: conceptualization, funding acquisition, investigation, methodology, supervision, validation, visualization, writing—review and editing.

Both authors gave final approval for publication and agreed to be held accountable for the work performed therein.

Conflict of interest declaration. We declare we have no competing interests.

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